# Technology Stack

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| **Date** | 15 March 2026 |
| **Team ID** | [Your Team ID] |
| **Project Name** | Streamline IT Procurement: Automating Standard Laptop Orders with Workflow Automation |
| **Maximum Marks** | 2 Marks |

## Define Your Technology Stack:

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs that you will use to build your solution. A well-chosen technology stack ensures that your application is scalable, maintainable, and aligned with your project requirements.

Fill out the table below to detail the specific technologies chosen for each component of your architecture and provide a brief justification for your choice.

### Technology Stack Details

| S.No | Architecture Component / Layer | Technology Chosen | Justification / Purpose |
|---|---|---|---|
| 1 | **Frontend / Client-Side** | **React.js / Tailwind CSS** | Provides a responsive, user-friendly portal for employees to view available laptop models, submit order requests, and track real-time approval status with component reusability. |
| 2 | **Backend / Server-Side** *(e.g., Node.js, Python/Django, Java)* | **Node.js (Express.js)** | Enables fast, asynchronous API processing to handle automated multi-level approval workflows and orchestrate requests between department managers and IT admins. |
| 3 | **Database / Data Storage** *(e.g., MySQL, MongoDB)* | **PostgreSQL** | Ensures strong relational data integrity, structured tables for laptop inventory management, employee records, purchase orders, and complete audit logging. |
| 4 | **Cloud / Hosting / Deployment** *(e.g., AWS, Azure, Heroku, Docker)* | **AWS (EC2, S3, Docker)** | Provides easy scalability, highly secure cloud storage for procurement documents/invoices, and containerized deployment for cost-effective hosting. |
| 5 | **Version Control & CI/CD** *(e.g., GitHub, GitLab, Jenkins)* | **GitHub & GitHub Actions** | Enables seamless team collaboration, structured code reviews, and automated testing/deployment pipelines for smooth updates. |
| 6 | **Third-Party APIs / Other Tools** *(e.g., Stripe, Twilio, Google Maps)* | **SendGrid API & Microsoft Teams / Slack Webhooks** | Sends automated email updates and real-time interactive notifications to department managers for instant order approval and delivery tracking. |